

# SHUCAYB AHMED

## Software Developer | Data & Network Engineering Student

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### PROFESSIONAL SUMMARY

Data and Network Engineering student at University West with hands-on project experience in software development, backend systems, networking, applied machine learning and embedded systems. I mainly work with Python, C++, JavaScript and SQL. I am seeking an internship or junior software role where I can contribute, ask questions and continue learning.

### TECHNICAL SKILLS

**Programming:** Python, C++, JavaScript, SQL, C

**Backend & Databases:** FastAPI, REST APIs, JWT, Argon2, MySQL

**AI / Machine Learning:** pandas, NumPy, scikit-learn, Random Forest, regression, classification, clustering, model evaluation

**Networking & Systems:** TCP/IP, Cisco Networking, ns-3, Linux, Windows, macOS

**Embedded Systems:** ESP32-C6, GPIO, I2C, interrupts, sensor integration

**Tools:** Git, GitHub, Google Colab, GNUplot, VS Code

### SELECTED PROJECTS

#### Aurora Grand Hotels | Full-stack team project

- Contributed to a hotel booking system with a client-server architecture, REST API and MySQL database.
- Worked on API integration, JWT authentication, Argon2 hashing, booking flow and server-side price calculation.

**Technologies:** Python, FastAPI, MySQL, JavaScript, HTML, CSS, REST API, JWT, Argon2

#### VPN vs Non-VPN Network Traffic Classification | AI + Networking

- Built a binary-classification pipeline for ISCXVPN2016 flows using nine numerical traffic measurements.
- Compared a baseline, SGDClassifier and Random Forest with three-fold stratified cross-validation; Random Forest was selected.
- Test results: 90.77% accuracy, 90.10% VPN precision, 92.33% VPN recall, 0.9120 VPN F1 and 0.9678 ROC-AUC.

**Technologies:** Python, pandas, NumPy, scikit-learn, Matplotlib, Google Colab

#### TCP Congestion Control Simulation - ns-3 | Network simulation

- Configured ns-3 simulations in C++ for TCP Cubic and NewReno.
- Collected congestion-window data and generated comparison graphs with GNUplot.

**Technologies:** C++, ns-3, TCP/IP, Linux, GNUplot

### ADDITIONAL TECHNICAL PROJECTS

#### TAM400 - Applied AI & Machine Learning Labs | University coursework

- Implemented BFS, DFS, A\* and hill climbing in Python.
- Worked with data preparation, regression, MNIST classification, K-Means and DBSCAN.

**Technologies:** Python, NumPy, pandas, scikit-learn, Matplotlib, Keras

#### Serial Bus Device Identification | In-progress system design project

- Evaluating software approaches for identifying and monitoring multiple LED devices on a shared serial bus.

**Technologies:** C++, serial communication, device monitoring

#### ESP32-C6 Event-Driven System | Embedded systems

- Worked with event-driven logic for button, IMU and touch events without blocking delays.
- Used GPIO, interrupts and I2C-based sensor communication.

**Technologies:** C/C++, ESP32-C6, GPIO, I2C, interrupts, IMU

## **EDUCATION**

**Data and Network Engineering | University West | 2024-present**

Trollhättan. Programming, computer networks, databases, distributed systems, AI and embedded systems.

**Technology & IT / Programming | Amerikanska Gymnasiet | 2021-2024**

Gothenburg.

## **CERTIFICATION**

CCNA: Introduction to Networks | Cisco Networking Academy | Verified through Credly

## **ADDITIONAL EXPERIENCE**

BASHUF | Swedish Young Enterprise

- Planned and delivered a student-run project with others, from initial idea to execution.

## **LANGUAGES**

Somali - native | Swedish - fluent | English - fluent